Python Programming - Module 1

Coding Practice Questions (No Answers)

Practice these on your own in IDLE / VS Code / any Python editor. Answers are intentionally not provided. Run your program with different inputs and compare against the sample output where given.

Module 1.1 - Basics, Input/Output, Operators

Level 1 - Warm-up

Q1. Write a program to read a name and age from the user and print: 'Hello <name>, you will be <age+1> next year.'

Q2. Write a program to read two integers and print their sum, difference, product, quotient (/), floor division (//), remainder and power.

Q3. Read a temperature in Celsius and convert it to Fahrenheit (F = C*9/5 + 32). Print it with 2 decimal places.

Q4. Read the radius of a circle and print its area and circumference (use 3.14159).

Q5. Swap two numbers (a) using a temporary variable and (b) without a temporary variable.

Q6. Read three numbers on one line separated by spaces and print their average. (Hint: split() and map()).

Q7. Use print() with sep and end so that the output is exactly: 10-20-30*

Q8. Read a number as a string and print its type, then convert it to int and float and print type and value of each.

Level 2 - Practice

Q9. Read the total seconds and display it as hours, minutes and seconds (e.g. 3725 -> 1 hr 2 min 5 sec).

Q10. Read a 3-digit number and print the sum of its digits using // and % operators only.

Q11. Create two lists a and b with the same content and a third name c = a. Print the results of a == b, a is b and c is a, and explain the difference using id().

Q12. Write a program that reads an integer and prints whether it is even or odd and whether it is positive, negative or zero.

Q13. Read the three sides of a triangle and display whether it is equilateral, isosceles or scalene (check validity first).

Q14. Evaluate and print the result of these expressions after predicting them on paper: 2 + 3 * 4 ** 2 // 5, (2 + 3) * 4 ** 2 // 5, 10 - 4 - 2, 2 ** 3 ** 2, not 0 or 0 and 1.

Q15. Print the type() of each value: 10, 10.0, '10', True, None, 3+4j, [1], (1,), {1}, {1: 1}, frozenset([1]).

Q16. Write a program to display a simple bill: read item name, price and quantity; print a neatly aligned line using f-string width specifiers, and the total with 2 decimals.

Q17. Check whether a given year is a leap year.

Q18. Read a number and print its multiplication table up to 10 in the format 5 x 3 = 15.

Module 1.2 - Strings

Level 1 - Warm-up

Q19. Read a string and print its length, first character, last character and the string in reverse (using slicing).

Q20. Print the string in uppercase, lowercase, title case and swapcase.

Q21. Read a string and count the number of vowels and consonants in it.

Q22. Check whether a string is a palindrome (ignore case).

Q23. Read a sentence and print the number of words in it.

Q24. Print every second character of a string, and the last 3 characters using negative slicing.

Q25. Replace all spaces in a string with underscores, without using replace() and then again using replace().

Q26. Read a string and print the number of uppercase letters, lowercase letters, digits and special characters.

Level 2 - Practice

Q27. Read a sentence and print each word on a new line along with its length.

Q28. Read a sentence and print the longest word in it.

Q29. Read a sentence and capitalise the first letter of each word without using title().

Q30. Remove all duplicate characters from a string, keeping the first occurrence order. (Input 'programming' -> 'progamin')

Q31. Read two strings and check whether they are anagrams.

Q32. Count the frequency of each character in a string and print it.

Q33. Read a string and toggle its case (lower to upper and upper to lower) without using swapcase().

Q34. Read a string and print it with the first and last characters exchanged.

Q35. Read a name in the form 'First Last' and print it as 'Last, First' using split() and join() / f-string.

Q36. Input a string such as 'a1b2c3' and print separately the letters and the digits, and the sum of the digits.

Q37. Check whether a string contains only digits, only alphabets, or both (alphanumeric), using string methods.

Q38. Read a string and a substring; print how many times the substring occurs and the index of every occurrence.

Q39. Print the following using one f-string/format call for each row, aligned in columns: Name (left, width 10), Marks (right, width 5), Percentage (2 decimals).

Module 1.2 - Lists

Level 1 - Warm-up

Q40. Create a list of 5 numbers from user input. Print the list, its length, max, min and sum.

Q41. Print the first, last and middle element of a list, and the list reversed (three different ways).

Q42. Append an element, insert an element at index 2, extend with another list, and then print the list after each step.

Q43. Remove an element by value and another by index (pop). Show what happens if the element is absent (handle with if / in).

Q44. Sort a list in ascending and descending order, once with sort() and once with sorted().

Q45. Find the second largest number in a list without using sort().

Q46. Count how many times a given element occurs in a list and find its first index.

Q47. Use del to remove the first element, then a slice of two elements, and finally the entire list.

Level 2 - Practice

Q48. Read a list of numbers and separate it into two lists: even numbers and odd numbers.

Q49. Remove duplicates from a list while maintaining the original order.

Q50. Merge two sorted lists into a single sorted list without using sort().

Q51. Rotate a list by k positions to the left (using slicing).

Q52. Find the sum of all the elements at even indices and odd indices separately.

Q53. Create a new list containing the squares of all the numbers in an existing list (use a for loop and then a list comprehension).

Q54. Read a list of names and print them sorted by length (shortest first).

Q55. Check whether a list is sorted in ascending order.

Q56. Flatten a nested list such as [[1, 2], [3, 4], [5]] into [1, 2, 3, 4, 5].

Q57. Find the common elements of two lists and the elements present in only one of them.

Q58. Create a 3x3 matrix using nested lists, read values from the user, print it row-wise and print the sum of the diagonal.

Q59. Given a list of integers, print the largest and smallest numbers along with their positions.

Q60. Demonstrate the difference between M = L and M = L.copy() by modifying one list and printing both.

Q61. Read a list of marks and print the number of students who scored above the average.

Module 1.3 - Dictionaries

Level 1 - Warm-up

Q62. Create a dictionary of a student with keys name, roll, and marks. Print it, then add a key 'city', modify marks, and delete roll.

Q63. Print all keys, all values and all key-value pairs of a dictionary using keys(), values() and items().

Q64. Read a key from the user and display its value, or 'Not found' if the key does not exist (use get() and also 'in').

Q65. Create a dictionary from two lists, keys and values, using a for loop and then using zip().

Q66. Merge two dictionaries using update(). What happens to common keys?

Q67. Print the length of a dictionary, its largest key and its smallest key.

Q68. Use pop(), popitem() and clear() on a dictionary and print the dictionary after each step.

Level 2 - Practice

Q69. Count the frequency of each word in a sentence using a dictionary.

Q70. Count the frequency of each character in a string and print the characters in sorted order of their keys.

Q71. Read n students' names and marks into a dictionary; print the name of the topper and the average marks.

Q72. Invert a dictionary: swap keys and values (assume values are unique).

Q73. Given a dictionary of item:price, print items costing more than 100, and the total cost of all items.

Q74. Sort a dictionary by its values in descending order and display it.

Q75. Create a dictionary where keys are numbers 1 to n and values are their squares.

Q76. Build a simple phone book: add, search, update and delete a contact using a menu-driven program with while loop.

Q77. Given a dictionary of dictionaries (student -> {subject: marks}), print the total marks for each student.

Q78. Group words by their first letter: ['apple','bat','ant','bag','cat'] -> {'a': [...], 'b': [...], 'c': [...]} using setdefault().

Q79. Check whether two dictionaries have the same keys and whether they are equal.

Q80. Use fromkeys() to create a dictionary where each key in a list is initialised to 0.

Module 1.4 - Tuples

Level 1 - Warm-up

Q81. Create a tuple of 5 numbers from user input and print its length, max, min and sum.

Q82. Print the first, last and reversed tuple using indexing and slicing.

Q83. Create a single-element tuple and print its type. Show what happens without the comma.

Q84. Use count() and index() on a tuple.

Q85. Concatenate two tuples and repeat a tuple three times.

Q86. Unpack a tuple (10, 20, 30) into three variables and print them.

Q87. Convert a tuple to a list, add an element, and convert it back to a tuple.

Level 2 - Practice

Q88. Find the largest and smallest elements of a tuple without using max() and min().

Q89. Given a tuple of numbers, create two tuples: one with even and another with odd numbers.

Q90. Remove a given element from a tuple (hint: convert to list).

Q91. Given a list of tuples [(1,'a'), (3,'c'), (2,'b')], sort it by the first element, then by the second element in descending order.

Q92. Given two lists names and marks, use zip() to print each student's name with marks and create a dictionary from them.

Q93. Use zip(*) to unzip [(1,'a'), (2,'b'), (3,'c')] into two separate tuples.

Q94. Convert a dictionary into a list of tuples using items(), then convert it back to a dictionary.

Q95. Store coordinates as tuples in a list and find the point farthest from the origin.

Q96. Read a tuple of names and print the names that start with a vowel.

Q97. Swap the values of two tuple variables a and b, and demonstrate nested tuples access: t = ((1, 2), (3, 4), (5, 6)) -> print 4.

Q98. Use zip() to add corresponding elements of two lists.

Q99. Find the index and frequency of every distinct element in a tuple.

Module 1.4 - Sets and Frozenset

Level 1 - Warm-up

Q100. Create a set from the list [1, 2, 2, 3, 3, 3] and print it. What do you observe?

Q101. Add one element using add() and many elements using update(); remove an element using remove() and discard().

Q102. Given A = {1, 2, 3, 4} and B = {3, 4, 5, 6}, print union, intersection, difference (A-B, B-A) and symmetric difference using both methods and operators.

Q103. Check whether one set is a subset / superset of another and whether two sets are disjoint.

Q104. Traverse a set using a for loop, and then in sorted order.

Q105. Create an empty set correctly and print its type. Also print the type of {}.

Q106. Create a frozenset from a list and try to add an element to it. Observe the result.

Level 2 - Practice

Q107. Remove duplicate elements from a list using a set. Print the unique count.

Q108. Read a sentence and print the unique words in it and count of unique words.

Q109. Read two strings and print the common characters, and characters present in only one of them.

Q110. Check whether a string is a pangram (contains all letters a-z) using sets.

Q111. Given three sets of students enrolled in Maths, Physics and Chemistry, find: students in all three, in only Maths, and in Maths or Physics but not Chemistry.

Q112. Find the elements that are present in exactly one of two lists.

Q113. Use frozenset to build a dictionary with frozenset keys, and show that {1,2} and {2,1} refer to the same key.

Q114. Check whether all elements of a list are unique using set and len().

Q115. Write a program to find the first repeating element in a list using a set.

Q116. Read two lists of numbers and print the union, intersection and the elements common to both in sorted order.

Mixed / Exam-Style Problems

Q117. Read a sentence. Print: number of words, number of unique words (set), frequency of each word (dict), and the words sorted alphabetically (list).

Q118. Read marks of 5 subjects for n students. Store each student as a tuple (name, [marks]) in a list; print each student's total and the rank list in descending order of total.

Q119. Given a list of integers, print a dictionary with keys 'even' and 'odd' containing lists of respective numbers.

Q120. Given a string, print a dictionary mapping each vowel to its count, and a set of consonants that appear in it.

Q121. Create a menu-driven program: 1. Add item to list 2. Remove item 3. Display sorted list 4. Search item 5. Exit.

Q122. Read emails separated by commas, split them into a list, remove duplicates (set), and print usernames (text before @) using slicing/split.

Q123. Write a program that reads a string of digits and prints the digit that appears the most number of times.

Q124. Create a list of tuples with (name, age); print the oldest and the youngest person, and the average age.

Q125. Given two lists of equal length, create a dictionary using zip(); then invert it; then print keys sorted by values.

Q126. Read a number n and generate a dictionary of {i: i2} for i in 1..n, then print only the entries whose squares are even.**

Predict the Output (run to verify afterwards)

```
# P1
s = "Python Programming"
print(s[7:], s[::-3], s.count("m"), s.find("o", 5))

# P2
L = [5, 3, 8, 1]
M = L
M.sort()
print(L, L is M)
N = sorted(L, reverse=True)
print(N, L)

# P3
d = {"a": 1, "b": 2}
d2 = d
d2["c"] = 3
print(len(d), d.get("z"), list(d.items())[-1])

# P4
t = (1, [2, 3], 4)
t[1].append(5)
print(t, len(t), t.count(4))

# P5
A = {1, 2, 3}
B = {3, 4}
print(A | B, A & B, A - B, A ^ B)
print(type({}), type(set()), type((1,)), type((1)))

# P6
print(list(zip([1, 2, 3], "ab")), dict(zip("xy", [10, 20])))
```